

STAT 314500 - Applied Partial Differential Equations
Assignment 5: Sturm-Liouville Eigenvalue Problems

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapter 5 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Show that $\lambda \geq 0$ for the eigenvalue problem

$$\frac{d^2\phi}{dx^2} + (\lambda - x^2)\phi = 0, \quad \phi(0) = \phi(1) = 0.$$

Is $\lambda = 0$ an eigenvalue?

(10 points)

Question 2 Consider the heat equation

$$\begin{aligned} c(x)\rho(x)\frac{\partial u}{\partial t} &= \frac{\partial}{\partial x}\left(K(x)\frac{\partial u}{\partial x}\right) + \alpha(x)u \\ u(0, t) &= 0 \\ u(L, t) &= 0 \\ u(x, 0) &= f(x), \end{aligned}$$

where c, ρ, K, α and f are smooth functions of x in $[0, L]$ with $c, \rho, K > 0$. Reduce this to an eigenvalue problem using separation of variables. Assuming that the eigenfunctions are known, when are the eigenvalues positive (i.e, the temporal modes are decaying exponentials)? Use this to solve the time dependent problem. What is the steady state solution (if it exists)?

(10 points)

Question 3 Consider the wave equation

$$\begin{aligned} \rho(x)\frac{\partial^2 u}{\partial t^2} &= \frac{\partial^2 u}{\partial x^2} + \alpha(x)u \\ u(0, t) &= 0 \\ u(L, t) &= 0 \\ u(x, 0) &= f(x) \\ \partial_t u(x, 0) &= g(x), \end{aligned}$$

where $\rho(x) > 0$ and $\alpha(x) < 0$. Reduce this to an eigenvalue problem using separation of variables. Assuming the eigenfunctions are known, solve the initial value problem.

(10 points)

Question 4 Consider the differential operator

$$\mathcal{L} = p(x)\frac{d^2}{dx^2} + r(x)\frac{d}{dx} + q(x).$$

1. By repeated integration by parts, find the adjoint operator $\mathcal{L}^*$ such that

$$\int_a^b [u\mathcal{L}^*(v) - v\mathcal{L}(u)]dx = H(x)|_a^b,$$

for some appropriate H . When is $\mathcal{L} = \mathcal{L}^*$ (self-adjoint)? (Note: $\mathcal{L}^*$ does not indicate a complex conjugation in this setting.)

2. Now suppose $u(a) = 0, u(b) + u'(b) = 0$. What should be the boundary conditions imposed on v at $x = a$ and $x = b$ so that $H(x)|_a^b = 0$ (called the adjoint boundary conditions)?
3. Let u be a solution to $\mathcal{L}(u) = f(x)$ with homogeneous boundary conditions ($\mathcal{L}$ is not necessarily self-adjoint). Show that such a solution exists only if f is orthogonal to all the solutions to the corresponding homogeneous adjoint problem. (Note: This is called the Fredholm alternative.)

Hint: The adjoint operator should be of the form $\mathcal{L}^* = p \frac{d^2}{dx^2} + \left(2 \frac{dp}{dx} - r\right) \frac{d}{dx} + \left(\frac{d^2 p}{dx^2} - \frac{dr}{dx} + q\right)$ (20 points)